

PH 582: Assignment-3

Topological Hall Effect in Magnetic Skyrmions

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Note: Parameters are taken in arbitrary units for numerical solution. Unless stated otherwise, parameters are set to: $t = 1, J_H = 1, m = 1, \gamma = 0$. If the question did not ask to plot for various parameters, a single set of parameters as stated in the report will have been used for the plot. Jupyter notebook was used for plotting using python and markdown cells can be used to find the code for any plot. Also the following short forms might be used at various places, both in report and the code:

- NN: Nearest neighbor (t : Hopping parameter)
- BS: Band structure
- $\mathcal{C}$: Chern number
- THC: Topological Hall Conductivity
- BZ: Brillouin Zone

1. **Skyrmion Lattice Construction:** We embed a single skyrmion within a 4×4 periodic supercell. The center is chosen at

$$(x_c, y_c) = \left(\frac{L_x - 1}{2}, \frac{L_y - 1}{2} \right), \quad (1)$$

with $L_x = L_y = 4$.

For a lattice site (x, y) :

$$r = \sqrt{(x - x_c)^2 + (y - y_c)^2}, \quad \alpha = \tan^{-1} \left(\frac{y - y_c}{x - x_c} \right). \quad (2)$$

We adopt a radial profile:

$$\theta(r) = \begin{cases} \pi \left(1 - \frac{r}{R} \right), & r \leq R, \\ 0, & r > R, \end{cases} \quad (3)$$

with $R = 1$. Corresponding spin texture is shown in Fig. 1.

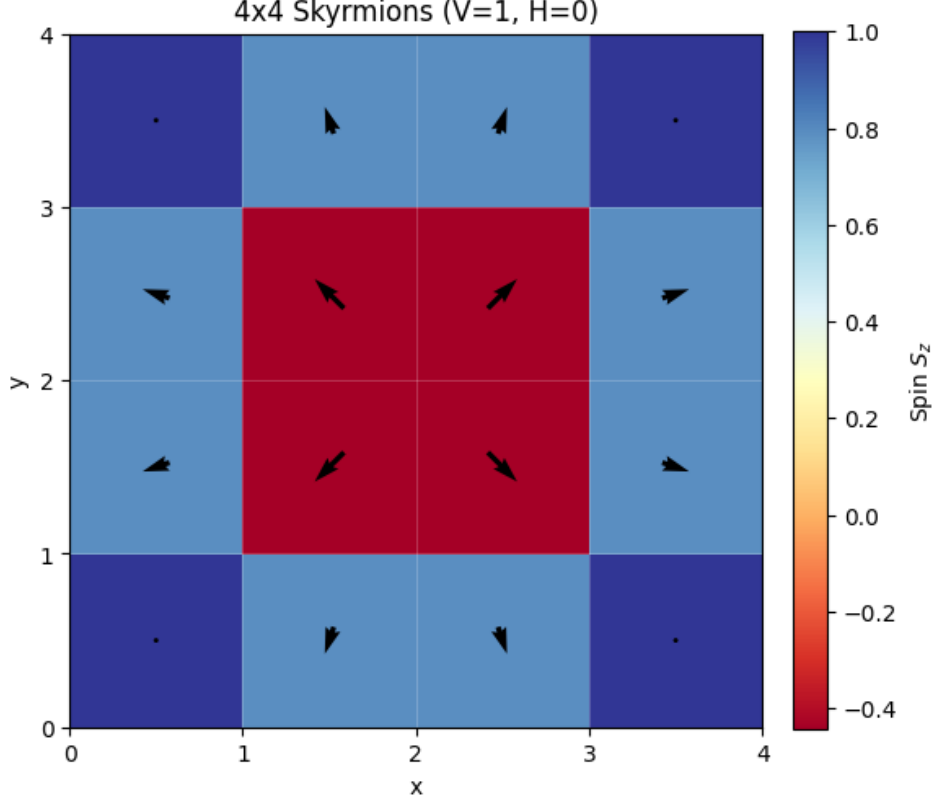


Figure 1: Magnetic Skyrmion spin texture on a 4x4 grid. Parameters taken are included in the text.

2. **Band Structure:** Due to the enlarged magnetic unit cell, the Brillouin zone is reduced:

$$-\frac{\pi}{4} \leq k_x, k_y < \frac{\pi}{4}. \quad (4)$$

The Hamiltonian at each $\mathbf{k}$ is a 16×16 matrix:

$$H(\mathbf{k}) |u_{n\mathbf{k}}\rangle = E_n(\mathbf{k}) |u_{n\mathbf{k}}\rangle. \quad (5)$$

Boundary-crossing hoppings acquire Bloch phases such as $e^{ik_x L_x}$ and $e^{ik_y L_y}$.

The Skyrmionic Hamiltonian is given as:

$$H = -t \sum_{\langle ij \rangle} c_i^\dagger c_j - J_H \sum_i \mathbf{n}_i \cdot (c_i^\dagger \boldsymbol{\sigma} c_i). \quad (6)$$

In the limit $J_H \gg t$, the electron spins align locally with $\mathbf{n}_i$. After projecting onto this low-energy sector, the Hamiltonian reduces to

$$H_{\text{eff}} = - \sum_{\langle ij \rangle} t_{ij}^{\text{eff}} d_i^\dagger d_j, \quad (7)$$

with effective hopping

$$t_{ij}^{\text{eff}} = t \left[\cos \frac{\theta_i}{2} \cos \frac{\theta_j}{2} + \sin \frac{\theta_i}{2} \sin \frac{\theta_j}{2} e^{-i(\phi_i - \phi_j)} \right]. \quad (8)$$

This expression can be recast as

$$t_{ij}^{\text{eff}} = t \cos \frac{\Theta_{ij}}{2} e^{ia_{ij}}, \quad (9)$$

where a_{ij} plays the role of an emergent gauge field responsible for the topological Hall effect.

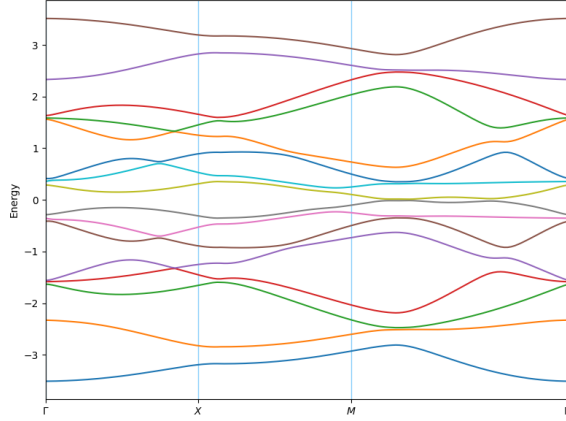


Figure 2: Full BS of 4x4 Spin Texture.

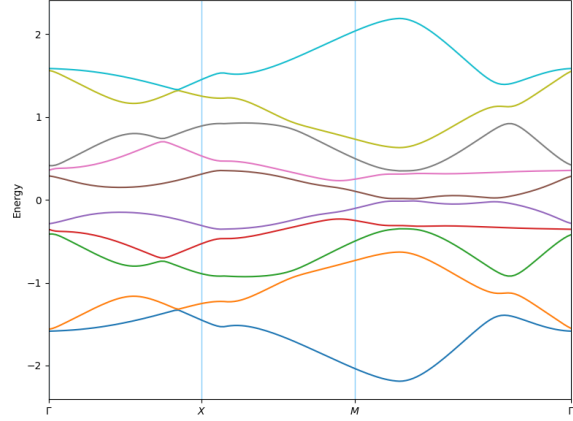


Figure 3: Bands near $E = 0$.

3. **THC:** We compute Berry curvature using a lattice discretization method. Define link variables:

$$U_x^{(n)}(\mathbf{k}_{ij}) = \frac{\langle u_{n,\mathbf{k}_{ij}} | u_{n,\mathbf{k}_{i+1,j}} \rangle}{|\langle u_{n,\mathbf{k}_{ij}} | u_{n,\mathbf{k}_{i+1,j}} \rangle|}, \quad (10)$$

$$U_y^{(n)}(\mathbf{k}_{ij}) = \frac{\langle u_{n,\mathbf{k}_{ij}} | u_{n,\mathbf{k}_{i,j+1}} \rangle}{|\langle u_{n,\mathbf{k}_{ij}} | u_{n,\mathbf{k}_{i,j+1}} \rangle|}. \quad (11)$$

The Berry flux per plaquette is:

$$F_{ij}^{(n)} = \arg \left[U_x^{(n)}(\mathbf{k}_{ij}) U_y^{(n)}(\mathbf{k}_{i+1,j}) (U_x^{(n)}(\mathbf{k}_{i,j+1}) U_y^{(n)}(\mathbf{k}_{ij}))^{-1} \right]. \quad (12)$$

The Chern number is obtained as:

$$C_n = \frac{1}{2\pi} \sum_{ij} F_{ij}^{(n)}. \quad (13)$$

The Hall conductivity at zero temperature is:

$$\sigma_{xy}(E_F) = \frac{e^2}{h} \sum_{E_n < E_F} C_n. \quad (14)$$

As the Fermi level crosses the bands, σ_{xy} exhibits quantized plateaus (refer Fig. 4).

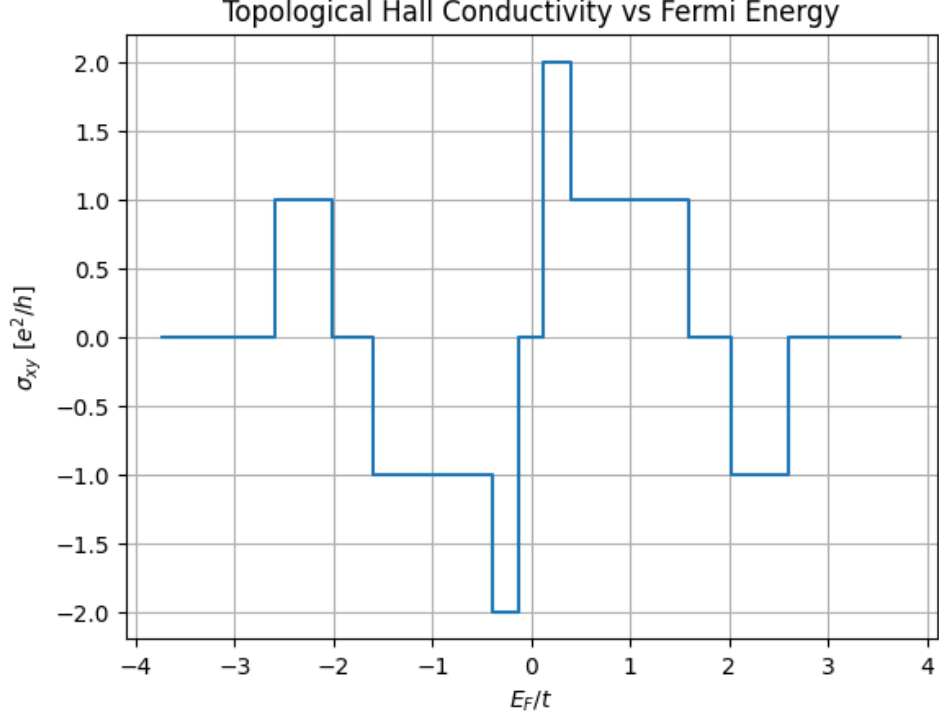


Figure 4: THC for 4x4 magnetic skyrmion.

4. THC Dependence on Vorticity and Helicity:

Since

$$\phi_i = m\alpha_i + \gamma, \quad (15)$$

only phase differences enter:

$$\phi_i - \phi_j = m(\alpha_i - \alpha_j), \quad (16)$$

leading to cancellation of γ .

Thus:

$$\sigma_{xy}(E_F; m, \gamma) = \sigma_{xy}(E_F; m). \quad (17)$$

Numerically:

$$\sigma_{xy}(m = -2) = -6 \frac{e^2}{h}, \quad (18)$$

$$\sigma_{xy}(m = -1) = +4 \frac{e^2}{h}, \quad (19)$$

$$\sigma_{xy}(m = +1) = -4 \frac{e^2}{h}, \quad (20)$$

$$\sigma_{xy}(m = +2) = +6 \frac{e^2}{h}. \quad (21)$$